
PROJECTS

Smart Biometric Attendance System

- Developed a complete biometric attendance solution for a company using ESP32, integrating a fingerprint device, web dashboard, and local data storage.
- Implemented employee check-in/check-out, holiday management, overtime calculation, attendance summaries, customizable settings, and long-term data storage of up to 12 months.

Firefighting Robot

- Developed an autonomous firefighting robot using IR sensors to detect fire/location conditions and initiate appropriate actions.
- Integrated sensor-based detection and automated movement for real-time response.

PID-Based Line Following Robot (LFR)

- Developed a PID-controlled line-following robot with an integrated display for real-time monitoring.
- Implemented configurable PID parameters with on-device editing and calibration for improved tracking accuracy and stability.

Robo Soccer Robot

- Built a remotely controlled soccer robot with adjustable speed and responsive control.
- Optimized motor control for fast movement, accurate direction changes, and competitive gameplay.

IoT-Enabled Environmental Monitoring Watch

- Developed a wearable IoT device using environmental sensors and Wi-Fi connectivity to monitor surrounding environmental conditions.
- Integrated sensor data collection and wireless communication for real-time monitoring.

Custom-Built Drone

- Designed, assembled, and tested a custom-built drone at university.
- Worked on hardware integration, flight control, testing, and performance optimization.

LEADERSHIP & ORGANIZATIONAL EXPERIENCE

Founding General Secretary | IEEE Manarat International University Student Branch | 2025–2026

- Co-founded and helped establish the IEEE Student Branch.
- Organized research seminars and technical workshops to promote research and engineering activities among students.

Executive Member | EEE Club | 2024–2025

- Assisted in organizing seminars, industrial visits, football and cricket tournaments, and student recreational activities.

Campus Secretary | Youth for Better Future (YBF) | 2024–2025

- Organized training sessions, training camps, sports tournaments, and blood donation programs.

IT & PR Secretary | Youth for Sustainable Future Society (YSF) | 2025–2026

- Developed and maintained websites for organizational activities.
- Organized IT training programs and supported digital communication and promotional activities.

Campus Ambassador | Phitron | 2025

- Promoted programming and competitive programming activities among university students.
- Encouraged students to participate in programming learning and skill-development activities.

Campus Ambassador | BEAR Summit, National Semiconductor Symposium

- Supported event organization and coordination.
- Assisted in engaging students and faculty members in a national-level semiconductor-focused event.

WORK EXPERIENCE

Technical Support & IT Assistant | Smart Solution | 2020–2021

- Provided computer hardware and software setup, troubleshooting, and technical support.
- Assisted with CCTV installation and configuration for government and private organizations.
- Supported system setup and maintenance across multiple client sites.

HONORS & ACHIEVEMENTS

- **Judge | Ismail Al-Jazari Robotics Fest** - Evaluated 300+ participants across multiple robotics competition categories.
- **Judge | Al-Khawarizmi Science Fest** - Evaluated 150+ project groups involving 600+ students.
- **5th Place | Rookie Code Clash Final** - Achieved 5th place in the final round and 18th place in the preliminary round.
- **700+ Programming Problems Solved** - Solved 700+ problems across competitive programming platforms including Codeforces, LeetCode, and HackerRank.

- **Online Leadership Training | Aspire Institute** - Completed leadership training featuring sessions conducted by Harvard Business School faculty.

COMPETITIONS & TECHNICAL ENGAGEMENTS

- **UIU CSE Fest** - Participated in Line Following Robot (LFR) competition.
- **NSU Tech Fest** - Participated in Robo Soccer competition.
- **BRAC University Traction** - Participated in Robo Soccer competition.
- **Cybernauts 2026** - Participated in Robo Soccer competition.

SEMINARS & CONFERENCES

- **Special One-Day Session on Modernizing Greater Dhaka's Transportation System | DTCA**
Shared ideas and proposed solutions for modernizing and improving the capital's transportation system.
- **Digital & Cyber Security Seminar | International Fire & Safety Security Expo 2022**
- **3rd Engineering Summit | Youth for Better Future (YBF)**

WORKSHOPS & TRAINING

- **Industrial Automation with PLC Technology** - AdvanTEK BD; hands-on experience with PLC-based industrial automation.
- **Antenna Design Using CST Microwave Studio** - Hands-on experience in antenna modeling and design using CST Microwave Studio.

INDUSTRIAL VISITS

- **Mirpur 132/33 kV Grid Substation** - Observed power transmission, grid infrastructure, and substation operations.
- **Energypac** - Visited transformer manufacturing facilities and observed transformer production processes.